INDIRECT SENSING FOR ROTOR FLUX POSITION OF PERMANENT MAGNET AC MOTORS OPERATING IN A WIDE SPEED RANGE

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Abstract – This paper describes an indirect sensing, or sensorless, method for the rotor flux position in Permanent Magnet (PM) ac motors operating in a wide speed range, while keeping maximum torque per current and/or maximum efficiency capabilities. The method of indirect sensing the rotor flux position proposed here is particularly applicable to trapezoidal back emf type of PM ac motors. For these types of motors, the open circuit stator phase voltages, also called back emf, have a trapezoidal shape that contains a fundamental and higher order frequency harmonics. In particular, the third harmonic component is extracted from the stator phase voltages while the fundamental and other polyphase components are eliminated via a simple summation of the three phase voltages. The resulting third harmonic signal keeps a constant phase relationship with the rotor flux for any speed and load condition, and is practically free of noises that can be introduced by the inverter switching, making this a very robust sensing method. Consequently, a minimum processing of the third harmonic signal is necessary in order to estimate the rotor flux position and to command the inverter current switching. In contrast with indirect sensing methods based on detection of the back-emf signal that require heavy filtering, the third harmonic signal needs only a small amount of filtering to eliminate the switching frequency and its side bands. As a result, the method proposed is not sensitive to filtering delays, allowing the motor to achieve a good performance for a wide speed range. Motor starting is also superior with this method since the third harmonic signal can be detected and processed at lower speeds than for the conventional method of back-emf sensing. Moreover, an alternative way to acquire the third harmonic signal without the need to access the stator neutral terminal is presented. This is particularly interesting when the motor neutral connection is not available or expensive to have access. The third harmonic indirect sensing scheme is implemented in the laboratory and compared to a conventional back-emf sensing method. The results of this comparison show that the method proposed is capable to drive a PM ac motor with higher torque per current and with higher efficiency within a wider speed range.

INTRODUCTION

A Permanent Magnet ac motor comprises a multi winding stator configuration, and permanent magnets mounted in either the rotor surface or in its interior. The magnets produce an air gap flux density distribution that is a function of the type of their magnetization characteristic and fabrication process. When they are magnetized axially a trapezoidal air gap flux density is produced, while when this magnetization is parallel to the magnet main axis a sinusoidal air gap flux density is generated. Because the main flux is produced by magnets that do not carry currents, the motor losses are restricted to the copper and iron losses in the stator and to iron loss in the rotor. Hence, this type of motor is suitable for applications where high efficiency is a concern. Inverter driven PM ac motors can operate in a self synchronous mode, where the rotor flux position commands directly the inverter current switching. In this case the motor is generally refereed to as Brushless dc motor (BLDC), Electronically Commutated Motor (ECM), or simply Brushless Permanent Magnet (BPM) motor. In order to obtain maximum torque per current, and also to guarantee stable operation, the motor is driven by an inverter so that the stator current vector is kept at quadrature with the rotor flux [1]. The work in Reference 1 also shows that for BPM motors operating at lower than rated output power, the maximum motor efficiency does not necessarily occur for a field oriented condition where the ratio torque per current is maximum. However, for operation at rated load condition, as is the case assumed in this paper, the maximum efficiency is achieved at maximum torque per current ratio.

The work reported in this paper concerns the trapezoidal back-emf motor type, supplied by an inverter that generates stator square wave currents with 120° conduction periods. Measurement of the rotor flux implies in the measurement of the rotor shaft position. A typical direct measurement utilizes encoders mounted in the shaft. In many applications, however, the use of direct position sensors is not desired due to reliability, cost, mechanical, and environmental constraints. On the other hand, several indirect rotor position sensing schemes have been proposed. The majority is based on the measurement of the open phase terminal voltage during the non-conduction period that corresponds to the air gap voltage or back-emf. Basically, all back-emf based schemes process the phase voltage via a low pass filter which has the function to eliminate high frequency noise and to introduce a phase delay for correct commutation of the stator current [2], [3], [4], and [5]. Figure 1 shows for one of the motor phases the current, and the terminal phase voltage

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before and after processing by a low pass filter. The wanted phase delay of $\pi/6$ introduced by the low pass filter is shown in the figure. The current is commutated at every zero crossing of the filtered signal $\langle v_{as} \rangle$, what guarantees that the current is in quadrature with the rotor flux. A problem with these schemes is its lack of robustness at low speed, since the back-emf amplitude is a direct function of the motor speed. Also, the delay introduced by the low pass filter is a function of the speed, making it not possible to keep the current vector in quadrature as speed changes. As a consequence, the torque per current and also the motor efficiency are adversely affected as the motor speed changes. Compensation for the filter phase delay is proposed in [9] with the introduction of a lead-lag network. That solution helps reducing the sensitivity of the phase delay with motor speed but still does not allow operation at wide speed range with maximum performance.

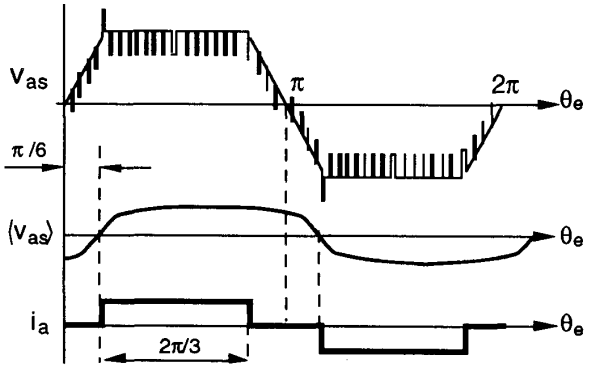


Fig. 1 Phase voltage and current for a BPM motor operating under zero crossing back-emf control method.

The method proposed here is based on the fact that in a symmetrical three phase Y-connected motor with trapezoidal air gap flux distribution, the summation of the three stator phase voltages results in an elimination of all polyphase components (fundamental and all the characteristics harmonics components like 5th, 7th, etc.); only the zero sequence components are left from the summation [10]. The resulting sum is dominated by the third harmonic component that keeps a constant phase displacement with the fundamental air gap voltage for any load and speed. An appropriate processing of the third harmonic signal allows the estimation of the rotor flux position and a proper inverter current control. In contrast with indirect sensing methods based on the back-emf signal, the resulting third harmonic signal is practically free of noise that can be introduced by the inverter switching; only a small amount of filtering is necessary to eliminate the switching frequency and its side bands. As a result, the method proposed is not sensitive to filtering delays, achieving a high performance for a wide speed range. A superior motor starting performance is also achieved because the third harmonic can be detected at low speeds. The next three sections cover the estimation of rotor flux position and current commutation control, the implementation of the proposed scheme, and experimental results obtained from a laboratory prototype.

ESTIMATION OF ROTOR FLUX POSITION AND CURRENT COMMUTATION CONTROL

Figure 2 shows the idealized air gap flux density distribution for a BPM motor with surface mounted magnets. The resultant trapezoidal air gap flux density has a dominant third harmonic component that links the stator phase windings inducing a third harmonic voltage component in each one of the phases. The summation of the three stator phase voltages results only in the third harmonic plus other high frequency components. This summation, however, requires access to the neutral point connection of the stator. This extra wire connection to the neutral carries only signal currents and although easy to install in the majority of applications it can, in some cases, represent extra cost or inconvenience to the motor installation. Two possible implementations for acquisition of the third harmonic voltage signal, one using the neutral connection (4-wire method) and another without neutral connection (3-wire method) are discussed.

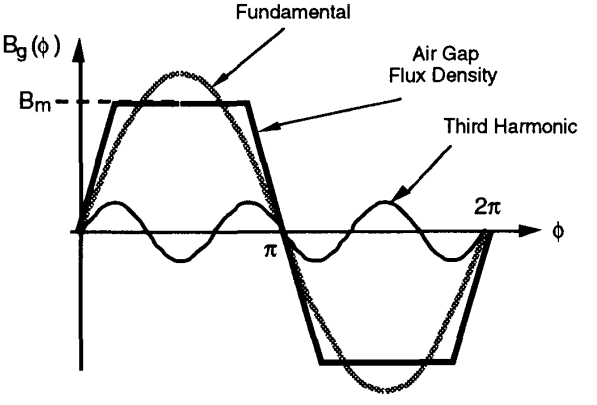


Fig. 2 Air gap flux distribution and its fundamental and third harmonic components.

The stator voltage equation for phase a, for instance, is written as,

$$v_{as} = r_s i_{as} + L_s \frac{d\, i_{as}}{dt} + e_{as} \qquad (1)$$

Similar expressions can be written for the other two stator phases. The stator phase resistance and inductance are

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represented by r_S and L_S respectively. The term e_{aS} represents the open phase voltage, or air gap voltage, also referred to as back-emf voltage. The harmonic content of the motor air gap or internal voltages e_{aS}, e_{bS}, and e_{cS}, is a function of the rotor magnets and stator winding configurations. For a full pitch magnet and full pitch stator phase winding, the internal voltage contains the following frequency components,

$$e_{as} = E(\cos \omega_e t + k_3 \cos 3\,\omega_e t + \ldots$$

$$k_5 \cos 5\,\omega_e t + k_7 \cos 7\,\omega_e t + \ldots) \quad (2)$$

Again, similar expressions can be written for the other two voltage terms. As mentioned above, these voltages are only a function of magnet rotor structure and stator winding, and stator teeth saturation does not affect them. Since no third harmonic currents circulate in a star connected stator, the third harmonic voltage component at the terminal voltage is only due to the air gap voltage, and consequently to the rotor flux. By adding up the three stator phase voltages we obtain,

$$v_{as} + v_{bs} + v_{cs} = 3\,E\,k_3 \cos 3\,\omega_e t + v_{high\,freq.} =$$

$$v_3 + v_{high\,freq.} \quad (3)$$

The rotor flux can be estimated from this third harmonic signal by integrating the resultant voltage v_3,

$$\lambda_{r3} = \int v_3\, dt \quad . \quad (4)$$

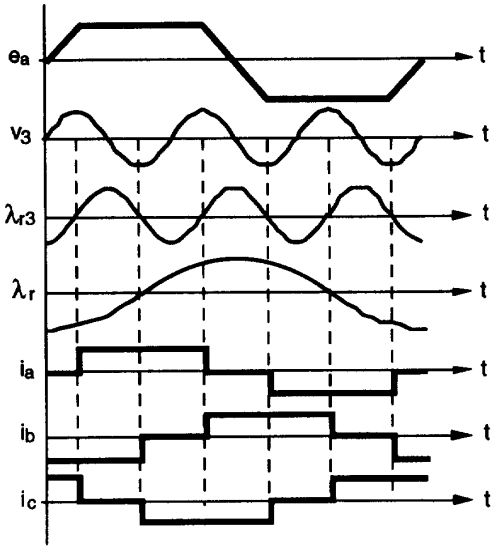


Fig. 3 BPM motor back-emf, third harmonic voltage, rotor flux and rotor flux fundamental components, and motor phase currents.

Figure 3 depicts the motor internal voltage corresponding to phase a, e_a, the third harmonic signal, v_3, obtained from the summation of the stator phase voltages, the rotor flux third harmonic component λ_{r3}, the rotor flux λ_r, and the stator phase currents. In order to obtain maximum torque per current, the stator current is kept at 90 electrical degrees with respect to the rotor flux. We note that the zero crossings of the rotor flux third harmonic component occur at 60 electrical degrees, exactly at every desired current commutation instant. Detection of the positive zero crossing of the voltage for phase a is necessary in order to implement a control algorithm that will command the correct stator currents. From figure 3, when the zero crossing of e_a is detected the control algorithm waits for the next zero crossing of the rotor flux third harmonic in order to turn the current in phase a on and to turn off the current in phase c. In the next zero crossing of λ_{r3} the control algorithm turns off current in phase b and turns on the current in phase c. At the next crossing of λ_{r3} , current in phase a is turned off while phase c is turned on. This procedure continues for the next three zero crossings of λ_{r3} signal, when an entire period is completed. It is important to mention that the zero crossing detection of the internal voltage (or air gap voltage) is possible because the phase current at that instant is zero. As shown in Fig. 3, this simple control algorithm allows the estimation of the position of the rotor flux and the control of the inverter currents so that the both motor current and rotor flux vectors are kept in quadrature.

This sensing method requires access to the neutral connection of the stator phases. It also requires a stator winding pole pitch and rotor pole pitch different of 2/3, otherwise the third harmonic rotor flux component does not link the stator winding and no third harmonic voltage is induced in the stator phases. The important advantage of this technique, besides its simplicity, is its low susceptibility to noise. The result of the summation of the three phase voltages contains the third harmonic signal and high frequency zero sequence components that can be easily eliminated by a low pass filter. This filtering action will not necessarily introduce a significant phase delay in the third harmonic signal which will be well below the filter cut off frequency. Signal detection at low speeds is possible because the third harmonic signal has a frequency three times higher than the fundamental back-emf, allowing operation in a wider speed range than techniques based on sensing the motor back-emf. The way the control scheme is implemented is the topic of the next section.

IMPLEMENTATION OF THE SENSING AND CONTROL SCHEMES

Figure 4 shows a generic three phase inverter used to drive, for instance, a three phase BPM motor. A star network of resistors of same value R is connected across the motor terminals and its common point labeled, n. Two other

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resistors, R_{dc}, are connected across the dc-link bus, forming a reference point labeled *h*. Note that the stator neutral point is labeled *s* and each one of the stator terminals labeled *a*, *b*, and *c*. The following voltage equations can then be derived:

$$v_{as} + v_{sn} + v_{na} = 0 \tag{5}$$

$$v_{bs} + v_{sn} + v_{nb} = 0 \tag{6}$$

$$v_{cs} + v_{sn} + v_{nc} = 0 \tag{7}$$

and when the three equations above are added up the result is,

$$(v_{as} + v_{bs} + v_{cs}) + 3v_{sn} + (v_{na} + v_{nb} + v_{nc}) = 0 \tag{8}$$

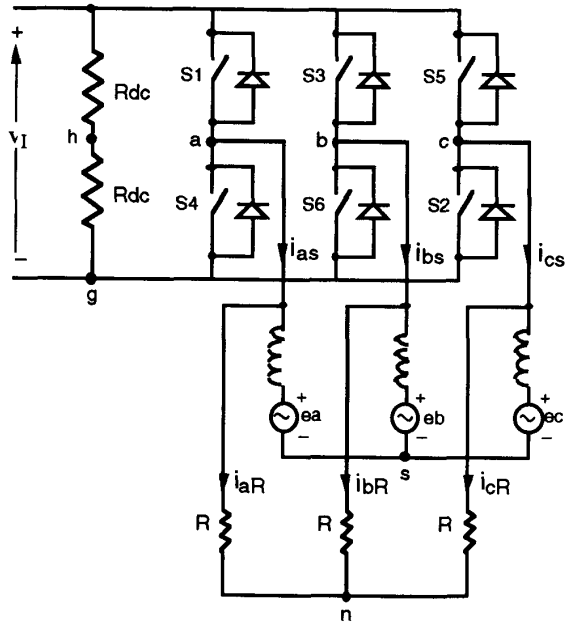


Fig. 4 Third harmonic sensing resistor interface connected to an inverter bridge.

Since the stator is star connected, the summation of the stator currents reduce to zero. Similarly, the summation of the three voltages across the star resistor network is zero, so that the summation above taking into account equation 3 reduces to,

$$v_{ns} = \frac{1}{3}\left[v_3 + v_{high\,freq.}\right] \tag{9}$$

It is interesting to note that the third harmonic voltage is obtained directly from the voltage across the two neutral points and no electronic summation means is necessary to add the three phase voltages. Furthermore, the problem of accessing the stator neutral with a fourth wire has also been solved and is discussed next. Referring back to Fig. 4, and using the label *g* for the negative rail of the dc link bus, one can write,

$$v_{ag} + v_{gn} + v_{ns} + v_{sa} = 0 \tag{10}$$

$$v_{bg} + v_{gn} + v_{ns} + v_{sb} = 0 \tag{11}$$

$$v_{cg} + v_{gn} + v_{ns} + v_{sc} = 0 \tag{12}$$

Let us assume that the motor is operating in self commutating mode where only two switches of the inverter bridge are conducting at any time. Assuming, for instance, that S_1 and S_2 are in conduction we have:

$$v_{ag} = v_I, \quad v_{cg} = 0, \quad v_{sc} = \frac{v_I}{2}, \quad \text{and} \quad v_{sa} = -\frac{v_I}{2} \tag{13}$$

where v_I represents the dc voltage input to the inverter bridge. After substituting these results in Eq. 12 and using the result from Eq. 9 we obtain,

$$v_{gn} = -v_{ns} - \frac{v_I}{2} = -\frac{1}{3}(v_3 + v_{high\,freq.}) - \frac{v_I}{2} \tag{14}$$

that shows that the voltage across the artificial neutral *n* and the negative rail of the dc link bus contains the third harmonic signal and the high frequency terms added to a dc level given by $v_I/2$. The same happens for other combinations of switches in conduction like, $S_2 - S_3$, $S_3 - S_4$, $S_4 - S_5$, $S_5 - S_6$, and $S_6 - S_1$, that correspond to a six step inverter operation sequence. For all of these switching combinations the voltage v_{gn} is given as in Eq. 14. The dc term, $v_I/2$ in Eq. 14 can be eliminated if we use point *h* as the reference instead of point *g*. Hence,

$$v_{hn} = -\frac{1}{3}(v_3 + v_{high\,freq.}) \tag{15}$$

During the inverter commutation, for instance when the switching sequence changes from *S*1–*S*2 to *S*2–*S*3, we have a condition where three switches are in conduction at the same time, in this case *S*2, *S*3 and the diode across *S*1. Consequently, we have that all inverter terminal voltages are constraint or defined as,

$$v_{ag} = v_I, \quad v_{bg} = v_I, \quad v_{cg} = 0 \tag{16}$$

Applying these values in equations 10 to 12 results in,

$$v_{hn} = -\frac{1}{3}(v_3 + v_{high\,freq.}) - \frac{1}{6}v_I \tag{17}$$

During a commutation in which two switches are connected to the negative rail of the dc link bus, for example during the commutation sequence *S*2–*S*3 to *S*2–*S*3–*S*4 to *S*3–*S*4, the voltage v_{hn} becomes,

$$v_{hn} = -\frac{1}{3}(v_3 + v_{high\,freq.}) + \frac{1}{6}v_I \tag{18}$$

Equations 17 and 18 show that an ac component varying from $-1/6\ v_I$ to $+1/6\ v_I$ is superimposed to the third

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harmonic signal. Since six commutations occur in any given period of the fundamental inverter output voltage, this superimposed component is at three times the fundamental frequency. This superimposed component introduces notches in the third harmonic signal and it is clear that they are at the same frequency as the third harmonic component for this condition where the motor is driven with a six step wave form (no PWM). The presence of the third harmonic component in the signal v_{hn} is not as clear when PWM is used because a commutation notch of $\pm 1/6\ v_I$ is generated at the PWM frequency.

In summary, the third harmonic stator voltage component can be obtained either from the voltage between the resistor network neutral and the motor stator neutral, v_{ns}, or from the resistor network neutral and the mid point reference at the dc bus, v_{hn}, in spite of the switching method used for the inverter bridge. A filter is necessary to eliminate the high switching frequency components. It is clear, then, that the stator third harmonic signal can be obtained without a direct access to stator neutral, eliminating the need of a fourth wire connection to the motor if desired. However, the signal v_{ns} is less noisy than v_{hn} since it does not present any notches due to the inverter commutations.

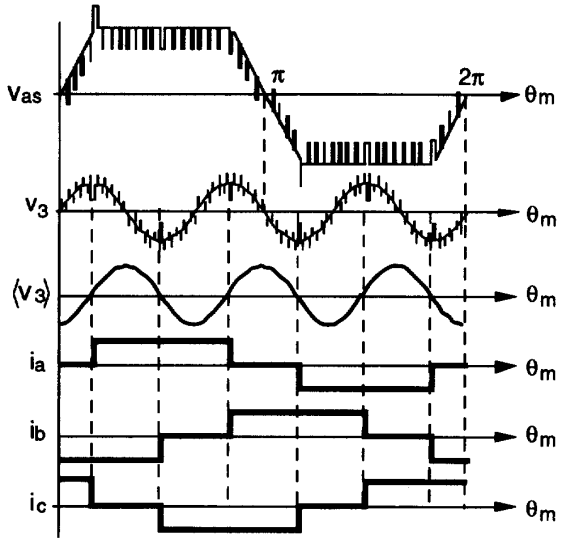


Fig. 5 Third harmonic voltage and its time integral used to control the inverter current commutations.

Figure 5 exemplifies how the third harmonic is processed and utilized to command the inverter currents commutation. Every zero crossing of the time integral of the third harmonic voltage signal is detected and counted by a microcomputer algorithm that keeps track of which inverter transistors to turn on or off. Note that the zero crossings of the integral of v_3 corresponds to a current switching instant. Detection of zero crossing of one of the phase voltages is also necessary in order to keep the synchronization of the third harmonic zeros crossing detection. Phase delay errors in the detection of the phase voltage are tolerated because the control algorithm uses that information only to keep track of the zero crossings of the third harmonic signal.

EXPERIMENTAL RESULTS

The proposed third harmonic based indirect rotor flux position sensor is implemented in the laboratory and compared to a conventional back-emf sensing scheme. Figure 6 shows in the upper trace the voltage between the motor stator and the resistor star network neutral points, v_{ns}, and in the bottom trace the voltage between the mid point in the dc link and the neutral point of the resistor star network, v_{hn}, for a 4-pole BPM motor driven by a six-step inverter and running at 1,400 rpm. The frequency spectra of that wave forms clearly show that the third harmonic component, in this case at a frequency of 140 Hz, has the largest amplitude. As explained in the previous section, a commutation notch at the same frequency as the third harmonic is present in the signal v_{hn} at every instant that two transistors and one diode of the inverter bridge are in conduction. It has been verified that the third harmonic stator voltage component keeps a constant phase relationship with the air gap voltage, and consequently with the rotor flux. The third harmonic voltage amplitude is directly proportional to the motor speed. For the prototype built we have a ratio of 4.3 mV/rpm for the third harmonic voltage amplitude.

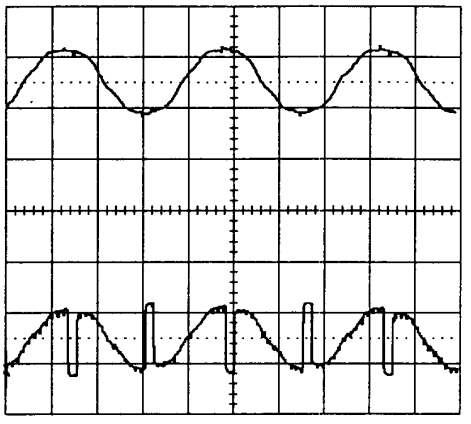


Fig. 6 Voltage signals v_{ns} (upper trace) and v_{hn} (bottom trace) for a BPM motor operating at 1,400 rpm and rated load. Scales: 5 volts/div. and 2 ms/div.

The control of the BPM motor using the method presented here is verified in the oscilloscope plot in Fig. 7. From top to bottom the plot shows: phase voltage, v_{as}, unfiltered third harmonic voltage signal, v_{ns}, motor phase c current, i_{cs}, and

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the inverter dc link current, i_{dc}. The motor is running at 1,800 rpm with twice rated load to verify the control robustness to an overload condition. As explained in the previous section, the plot shows that the motor phase current commutations occur at every zero crossing of the time integral of the third harmonic signal v_{ns}, or at every maximum/minimum of v_{ns}.

It has been verified experimentally that this control scheme is capable to operate the BPM motor from rated speed (6,000 rpm) down to about 100 rpm. At lower speeds detection of both the third harmonic and the zero crossing of the phase voltage becomes difficult to the lower signal levels. The back-emf control scheme used for comparison was able to drive the motor from 6,000 rpm to about 1,000 rpm.

Operation of the BPM motor using the third harmonic signal obtained from three wires only is shown in Fig. 8. From top to bottom the plot shows the third harmonic voltage signal v_{hn} , the time integral of v_{hn} after filtering, and the motor phase a current. As expected, a higher degree of noise is present in the signal v_{hn} which is eliminated with appropriate filtering.

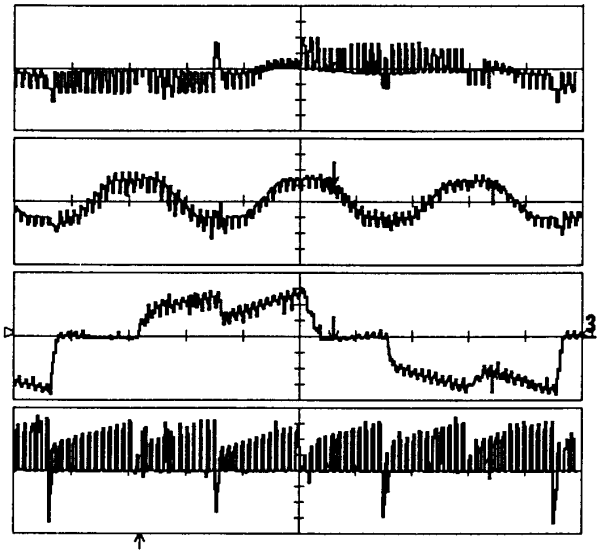


Fig. 7 Voltages and currents for steady state operation of a BPM motor running at 1,800 rpm and twice rated load. From top to bottom: motor phase a voltage (100 V/div.), third harmonic voltage signal (5 V/div.), motor phase c current (1 A/div.), and inverter dc link current (1 A/div.).

The starting performance of the BPM motor under the third harmonic control is evaluated and Fig. 9 shows the motor speed and motor phase current for a starting transient with twice rated load. During start, the motor control is open loop and the motor operates as a synchronous motor. After the motor reaches a certain speed, about 80 rpm for this case, the third harmonic signal is detected and the rotor flux position control loop is closed; from that point on the motor operates in self commutating mode.

Figure 10 compares the BPM motor efficiency as a function of speed with constant rated load for the two sensing methods. The BPM motor drive system was designed for maximum efficiency at rated load and at rated speed of about 6,000 rpm. At that point both sensing methods offer approximately the same efficiency. As expected, as the speed drops so does the BPM motor efficiency. However, the efficiency drop is more accentuated for the back-emf sensing scheme because the delay introduced by the low pass filter decreases with the motor speed.

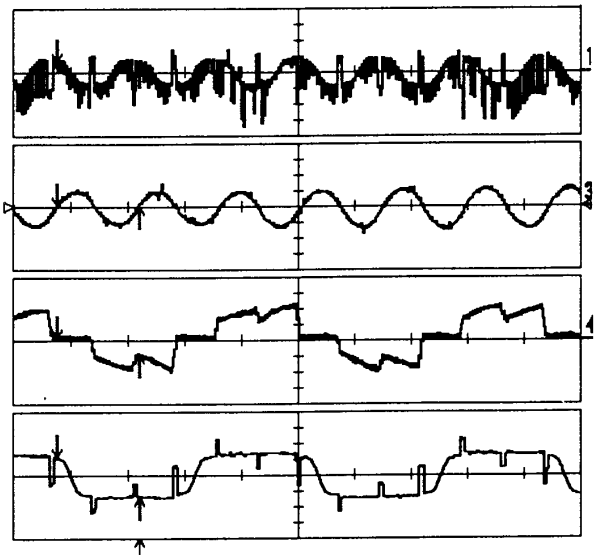


Fig. 8 Steady state operation of the BPM motor with third harmonic 3-wire method. From top to bottom: third harmonic signal v_{hn} (5 v/div.), time integral of v_{hn}, and motor phase a current (1 A/div.). Motor is operating at 1,800 rpm and rated load.

This phase delay introduced by the filter is responsible for the loss of field orientation, or in other words, loss of the quadrature condition between rotor flux and stator current. The immediate consequence is the reduction of the torque per current ratio of the motor, what implies in larger copper losses. The motor efficiency when controlled by the third harmonic method starts at the high speed end at approximately the same value as for the back emf method. However, the motor efficiency does not drop as much as the speed is reduced. Note also that the operating speed range is from 1,000 to 6,000 rpm for the back-emf scheme while it is from 100 to 6,000 rpm for the third harmonic method.

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In conclusion, these results show that the BPM motor control method based on the third harmonic signal as presented here is capable of keeping motor operation at maximum efficiency and maximum torque per current for a wider speed range than the conventional back-emf control method. Furthermore, the starting performance is enhanced when the proposed method is applied.

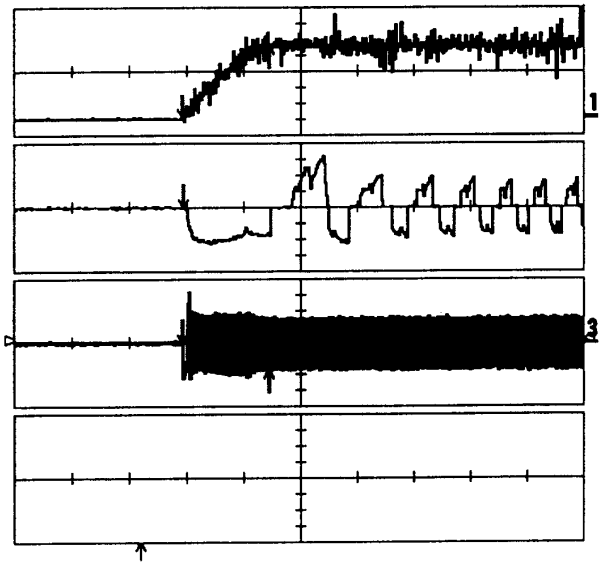


Fig. 9 Starting transient for the BPM motor with twice rated load. From top to bottom: motor speed (1,000 rpm/div.), time expanded motor current (2 A/div. – 0.1 s/div.), motor current (2 A/div.). Time scale: 2 s/div.

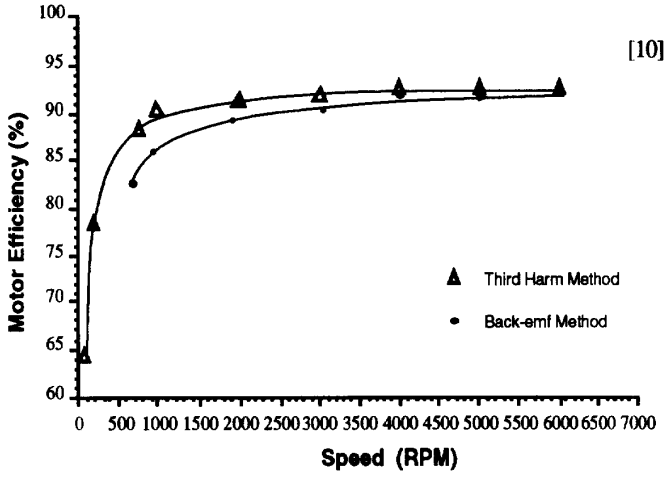


Fig. 10 Motor efficiency as a function of speed for constant rated load comparing the performance of the back-emf and the third harmonic commutation control methods.

ACKNOWLEDGMENTS

The author would like to acknowledge the contributions and support received to this work from Arne Nystuen, Don VanderMolen and Roger Nyland from Whirlpool Corporation.

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